1. Define IOC

IOC stands for **Indicator of Compromise**. It is a piece of evidence that shows a computer system or network may have been attacked or infected by malware. Security professionals use IOCs to detect, investigate, and respond to cyber threats before they cause further damage.

2. Give Examples

Some common examples of Indicators of Compromise (IOCs) are:

- Unusual IP addresses communicating with a device.
- Suspicious file hashes that match known malware.
- Unknown or malicious domain names.
- Unexpected changes to system files or registry settings.
- Multiple failed login attempts followed by a successful login.
- Unusual network traffic or data being sent outside the organization.
- Antivirus alerts or detection of malicious files.

These indicators help security teams identify possible security incidents and take appropriate action.

3. Explain Limits

Although IOCs are useful, they have some limitations:

- They usually identify attacks **after** they have already occurred.
- Attackers can change file names, IP addresses, or domains to avoid detection.
- Some IOCs may generate false positives, which can waste time during investigations.
- They do not always explain how the attack happened or what the attacker intends to do.
- New or unknown threats may not have existing IOCs, making them difficult to detect.

In conclusion, IOCs are an important part of cybersecurity because they help detect and investigate security incidents. However, they should be used together with other security measures, such as continuous monitoring and threat hunting, for better protection.